New chat

L6_S2_A.pdf
PDF

A scribe follows the following structure- Header information like course name, lecture name or number, topic and name of the instructor
Introduction and overview
Main body content
Important notes/ remarks
Conclusion

Using the same structure as the above mentioned example, generate a lecture scribe for the provided content and do not introduce new content.
The scribe must serve as exam reference material which includes detailed definitions and annotations, assumptions and conditions, statement of theorems or results, proofs or proof sketches, step by step, derivations and worked examples only if they appear in the provided context, with intermediate steps and clear logical flow between ideas.

Below is a **complete lecture scribe** generated **strictly from the provided PDF content**, following **exactly the structure you specified** and written as **exam-oriented reference material**.
No new concepts, examples, or results have been introduced beyond what appears in the slides and annotations.
(Page numbers are mentioned to help you trace content in the slides.)

Lecture Scribe

Header Information

- **Course Name:** CSE400 – Fundamentals of Probability in Computing
- **Lecture Number & Title:** Lecture 6 – Discrete Random Variables, Expectation and Problem Solving

- **Primary Topics Covered:**
  Random Variables, PMF, Bayes' Theorem, Independent Events, Types of Discrete Random Variables (Bernoulli, Binomial, Geometric, Poisson)
- **Instructor:** Dr. Dhaval Patel, PhD (Associate Professor, CSE)
- **Institution:** SEAS, Ahmedabad University, India
- **Date:** January 22, 2025

Introduction and Overview

This lecture develops the formal framework of **random variables**, focusing primarily on **discrete random variables** and their associated probability models. The lecture begins by defining random variables as functions on a sample space and motivates their use through visual and probabilistic interpretations.

Subsequently, the lecture introduces the **Probability Mass Function (PMF)** and establishes its fundamental properties. Using this foundation, **Bayes' Theorem** is revisited and applied to practical examples. The notion of **independent events** is formally defined and examined through communication network examples.

The lecture concludes with a systematic treatment of the **main classes of discrete random variables**—Bernoulli, Binomial, Geometric, and Poisson—along with their assumptions, PMFs, and applications.

Main Body Content

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1. Random Variables: Motivation and Definition

Definition:
A **random variable** X on a sample space Ω is a function

$$X : \Omega \to \mathrm{R}$$

that assigns a real number to each outcome $\omega \in \Omega$.

Although X maps into R, in this lecture we restrict attention to **discrete random variables**, meaning the set of values $\{X(\omega)\}$ is **finite or countably infinite**.

Interpretation:

- Outcomes in the sample space are abstract.
- A random variable converts outcomes into numerical values.
- Multiple outcomes may map to the same value of the random variable.

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2. Discrete vs Continuous Random Variables

Discrete Random Variables

- Countable support
- Probabilities assigned to **individual values**
- Characterized by a **Probability Mass Function (PMF)**

Continuous Random Variables

- Uncountable support
- Probabilities assigned to **intervals**
- Characterized by a **Probability Density Function (PDF)**

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3. Example: Tossing Three Fair Coins

Let Y be the number of heads obtained when three fair coins are tossed.

Possible values:

$$Y \in \{0, 1, 2, 3\}$$

Probabilities (computed by enumerating sample points):

$$P(Y = 0) = \frac{1}{8}$$
$$P(Y = 1) = \frac{3}{8}$$
$$P(Y = 2) = \frac{3}{8}$$
$$P(Y = 3) = \frac{1}{8}$$

Normalization condition:

$$\sum_{i=0}^{3} P(Y = i) = 1$$

This validates the legitimacy of the PMF.

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4. Probability Mass Function (PMF)

Definition:

Let X be a discrete random variable with range

$$R_X = \{x_1, x_2, x_3, \dots\}$$

The function

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X.

Key Property (Normalization):

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

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5. PMF Example with Parameter λ

Given:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

Using the identity:

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

Step-by-step derivation:

$$\sum_{i=0}^{\infty} p(i) = ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Thus,

$$P(X = 0) = e^{-\lambda}$$

For:

$$P(X > 2) = 1 - P(X \leqslant 2)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

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6. Bayes' Theorem

Bayes' Formula (Proposition 3.1):

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)P(B_j)}$$

Interpretation:

- $P(B_i)$: Prior probability
- $P(B_i \mid A)$: Posterior probability after observing event A

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7. Bayes' Theorem: Communication System Example

A binary communication system transmits 0 or 1 with errors.

Given:

$$P(0 \mid 0) = 0.95, \quad P(1 \mid 0) = 0.05$$
$$P(0 \mid 1) = 0.10, \quad P(1 \mid 1) = 0.90$$

Tasks solved in lecture:

- Compute $P(0 \text{ received})$, $P(1 \text{ received})$
- Compute posterior probabilities using Bayes' rule
- Compute **overall probability of error**:

$$P_e = P(\text{received} \neq \text{transmitted})$$

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8. Independent Events

Definition:
Two events A and B are independent if:

$$P(A \mid B) = P(A) \quad \text{and} \quad P(B \mid A) = P(B)$$

Equivalently:

$$P(A \cap B) = P(A)P(B)$$

Mutual Independence (Three Events):
Events A, B, C are mutually independent if:

$$P(A \cap B) = P(A)P(B)$$
$$P(A \cap C) = P(A)P(C)$$
$$P(B \cap C) = P(B)P(C)$$
$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

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9. Types of Discrete Random Variables

9.1 Bernoulli Random Variable

Experiment: Single trial with outcome Success or Failure.

Definition:

$$X = \begin{cases} 1, & \text{Success} \\ 0, & \text{Failure} \end{cases}$$

PMF:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p \quad (0 < p < 1)$$

9.2 Binomial Random Variable

Experiment:
n independent Bernoulli trials with success probability p.

Definition:
X = number of successes in n trials.

Notation: $X \sim B(n, p)$

PMF:

$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}, \quad i = 0, 1, \dots, n$$

9.3 Geometric Random Variable

Experiment:

Independent trials performed until first success.

Definition:

X = number of trials required for first success.

PMF:

$$P(X = n) = (1 - p)^{n-1} p, \quad n = 1, 2, \ldots$$

Normalization check:

$$\sum_{n=1}^{\infty} (1 - p)^{n-1} p = 1$$

9.4 Poisson Random Variable

Definition:

A random variable X is Poisson with parameter $\lambda > 0$ if:

$$P(X = i) = \frac{e^{-\lambda} \lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Normalization:

$$\sum_{i=0}^{\infty} P(X = i) = 1$$

Important Note:

Poisson distribution can approximate a Binomial $B(n, p)$ when:

- n is large
- p is small
- np is of moderate size

Important Notes / Remarks

- PMF validity requires non-negativity and unit total probability.
- Independence is stronger than pairwise independence.
- Bayes' Theorem updates beliefs using observed data.
- Poisson approximation is a key exam concept.

- Discrete RV models must match the experiment's structure.

Conclusion

This lecture establishes the theoretical backbone of **discrete probability modeling**. Starting from the abstract definition of random variables, it systematically develops PMFs, conditional probability, Bayes' Theorem, and independence. The lecture culminates in a structured taxonomy of discrete random variables, each tied to specific experimental assumptions and mathematical formulations.

These concepts form the foundation for **expectation, variance, and probabilistic analysis in computing systems**, making them essential for both exams and advanced applications.

If you want, I can:

- Convert this into **LaTeX exam notes**
- Split it into **unit-wise revision sheets**
- Create **solved numerical practice problems** strictly aligned with this lecture